

SwiftAssist AI Customer Support System

Product Management Weekly Project - W13

1. Problem Statement

Challenge:

SwiftAssist is experiencing a critical customer support bottleneck that is directly impacting customer satisfaction and retention.

Key Issues:

- Customers wait an average of **24 hours** for refund status updates, leading to frustration and higher churn
- Manual support agents are overwhelmed with repetitive queries (order status, delivery updates, refund checks)
- Support response times are consistently exceeding customer expectations
- High volume of basic queries is preventing human agents from focusing on complex issues that require empathy and problem-solving
- Customer satisfaction scores are declining due to delayed responses

Business Impact:

- Increased customer churn rate
- Rising operational costs due to expanding support team requirements
- Brand reputation damage in competitive food delivery market
- Lost revenue from dissatisfied customers switching to competitors

2. Proposed AI Workflow

Core AI Agent Process Flow:

Step 1: Input Processing

- Customer submits query via chat, email, or app
- AI performs natural language processing to categorize query type
- Extracts key information (order ID, phone number, complaint type)

Step 2: Data Retrieval & Reasoning

- Fetches relevant data from databases:** Order history and status, Payment and refund records, Delivery tracking information, Customer profile and previous interactions
- Applies business logic to determine appropriate response

Step 3: Decision Making

- Simple Queries (Auto-resolve):** Order status, delivery ETA, refund status
- Medium Complexity (Template + Data):** Delayed delivery explanations, standard refund processing
- Complex Issues (Human Escalation):** Food quality complaints, special refund cases, billing disputes

Step 4: Response Generation

- Generates personalized, empathetic response using customer data
- Provides specific information (order details, timelines, next steps)
- If escalating, summarizes context for human agent

Step 5: Follow-up & Learning

- Schedules follow-up if required
- Logs interaction for continuous learning
- Updates customer satisfaction metrics

3. Example Prompts/LLM Workflows

Prompt 1: Refund Status Inquiry

You are SwiftAssist's AI customer support agent. A customer is asking about their refund status. Context: Customer ordered on [DATE], order was cancelled, refund initiated. Instructions: 1. Check refund status in payment system 2. If processing: Provide expected completion date (3-5 business days from initiation) 3. If completed: Confirm refund amount and payment method 4. If delayed: Apologize and escalate to human agent 5. Maintain friendly, helpful tone Response should include: - Acknowledgment of concern - Specific refund details - Clear next steps or timeline - Contact information if further assistance needed

Prompt 2: Order Delivery Delay

You are handling a delivery delay complaint for SwiftAssist. Customer Data: Order # [ID], estimated delivery [TIME], current status [STATUS] Process: 1. Acknowledge the inconvenience caused 2. Provide real-time order tracking information 3. Calculate revised delivery estimate based on current location 4. Offer appropriate compensation (discount/refund) based on delay duration 5. If delay >60 minutes, automatically escalate to human agent Tone: Apologetic but solution-focused, avoid technical jargon

Prompt 3: General Order Status

Customer is asking "Where is my order?" Your role: Provide comprehensive order update for SwiftAssist. Steps: 1. Retrieve order details using customer phone/email 2. Present current status in clear, simple language 3. Provide estimated delivery time with 5-minute buffer 4. Share delivery person contact if order is out for delivery 5. Proactively address potential concerns (weather delays, restaurant preparation time) Format: Conversational, informative, and reassuring

4. Key Metrics (KPIs)

Primary Success Metrics:

Response Time Reduction

Target: Reduce average response time from 24 hours to under 5 minutes
Measurement: Time between customer query and first meaningful response
Success Threshold: 90% of queries resolved within 5 minutes

Automation Rate

Target: 70% of customer queries resolved without human intervention
Measurement: (AI-resolved queries / Total queries) × 100
Success Threshold: Maintain quality while achieving 70% automation

Customer Satisfaction (CSAT)

Target: Improve CSAT score from current baseline to 4.2/5.0
Measurement: Post-interaction customer ratings
Success Threshold: Maintain >4.0 rating with increased query volume

Secondary Metrics:

- First Contact Resolution Rate:** 85% target
- Escalation Rate:** <30% of total interactions
- Cost per Query:** 60% reduction in handling cost

5. Risks and Challenges

Risk 1: Incorrect Refund Processing

Problem: AI might approve refunds incorrectly, leading to financial losses

Solution:

- Implement approval thresholds: Human approval required for refunds >€500
- Create refund eligibility checklist with strict validation rules
- Regular audit of AI refund decisions with feedback loop

Risk 2: Lack of Emotional Intelligence

Problem: AI responses may seem robotic and fail to address customer frustration appropriately

Solution:

- Develop empathy-focused response templates
- Sentiment analysis to detect frustrated customers and escalate immediately
- Regular review of customer feedback to improve tone and messaging

Risk 3: Data Privacy and Security

Problem: AI system handling sensitive customer and payment information

Solution:

- Implement role-based data access controls
- Regular security audits and compliance checks
- Anonymize data used for AI training and testing

Risk 4: Over-reliance on Automation

Problem: Complex issues might be misclassified and handled inappropriately

Solution:

- Continuous monitoring of escalation patterns
- Regular human agent training on AI handoffs
- Feedback mechanism for customers to request human support at any time

6. AI Workflow Visualization



Key Performance Indicators:



Implementation Roadmap

Phase 1 (Month 1-2)

Foundation

- Develop core AI classification system
- Create response templates for top 10 query types
- Integrate with existing order and payment systems

Phase 2 (Month 3-4)

Expansion

- Add advanced NLP capabilities
- Implement sentiment analysis
- Launch beta with limited customer segment

Phase 3 (Month 5-6)

Optimization

- Full deployment across all customers
- Continuous learning implementation
- Advanced analytics and reporting dashboard

Success Criteria

- Immediate (3 months):** 50% reduction in response time, 40% automation rate
- Medium-term (6 months):** 70% automation rate, 4.0+ CSAT score
- Long-term (12 months):** 80% automation rate, 25% cost reduction, improved NPS scores

This AI-driven solution will transform SwiftAssist's customer support from a reactive cost center into a proactive competitive advantage, ensuring customers receive instant, accurate support while allowing human agents to focus on building relationships and solving complex problems.